The sealed statements of LocalityLean

These 60 statements are the ones that `lake comparator` checks. Each is followed by the name of its formal version in `verify/Challenge.lean`.

Nets of local algebras on a lattice

A net over a preordered set R of regions in a C*-algebra A assigns to each region O a star subalgebra $\mathcal{A}(O)$ of A. A C*-algebra is a unital complex C*-algebra, and a star subalgebra is a unital complex star subalgebra. On a lattice with site set Λ the regions are the subsets of Λ, ordered by inclusion. For elements a, b of A, $[a, b] = ab - ba$.

1. A net over R in a C*-algebra A assigns to each region O a star subalgebra $\mathcal{A}(O)$ of A; an assignment $O \mapsto F(O)$ is a net exactly when it is monotone ($O \leq O'$ implies $F(O) \subseteq F(O')$, isotony) and every $F(O)$ is norm-closed.
`Sealed.net_iff`

2. The lattice causal index on the subsets of a site set Λ: two regions are orthogonal exactly when they are disjoint, and a region depends on another exactly when it is contained in it. A net $\mathcal{A}$ is causal for this index exactly when elements of the algebras of disjoint regions commute: $X \cap Y = \emptyset$, $a \in \mathcal{A}(X)$ and $b \in \mathcal{A}(Y)$ give $ab = ba$.
`Sealed.lattice_index`

3. The commutator seminorm of a region. For a net $\mathcal{A}$ on the subsets of Λ, a region S and $c \in A$,

$$N_S(c) = \sup\{\|[c, b]\| : b \in \mathcal{A}(S), \|b\| \leq 1\};$$

this supremum is finite, with $0 \leq N_S(c) \leq 2\|c\|$ and $\|[c, b]\| \leq N_S(c)$ for every b in the unit ball of $\mathcal{A}(S)$. A set S crosses a region Y when it meets Y and is not contained in Y.
`Sealed.commutator_seminorm`

The CCR algebra

Let V be a real vector space and σ a real bilinear form on V. The Hilbert space $\ell^2(V)$ consists of the square-summable complex functions on the set V. A bilinear form is alternating when $\sigma(u, u) = 0$ for every u.

4. The Weyl operators and the CCR algebra: for $u \in V$ the Weyl operator $W(u)$ is the bounded operator on $\ell^2(V)$ with

$$(W(u)f)(v) = e^{i\sigma(u,v)/2} f(v + u);$$

the CCR algebra $\mathrm{CCR}(V, \sigma)$ is the norm closure of the star subalgebra generated by the Weyl operators, a C*-algebra; and $W(u)$, regarded as an element of $\mathrm{CCR}(V, \sigma)$, is the same operator.
`Sealed.weyl_and_CCR`

5. For alternating σ every Weyl operator is unitary, $W(u)^* = W(-u)$, and the Weyl relation

$$W(u)W(w) = e^{-i\sigma(u,w)/2}W(u+w)$$

holds.
`Sealed.weyl_relations`

Local generators

A family of local generators on a net $\mathcal{A}$ is indexed by a finite type ι; its i-th term has a finite support $S_i \subseteq \Lambda$, a bounded complex-linear map $L_i : A \to A$ and a strength κ_i. The norm of a bounded linear map of A is the operator norm, and e^T is the exponential series of a bounded linear map T.

6. A family of local generators: given finite supports S_i, bounded linear maps $L_i : A \to A$ and reals κ_i for $i \in \iota$, there is a family of local generators on $\mathcal{A}$ with these data exactly when (i) $\kappa_i \geq 0$; (G1) locality in the argument, $\|L_i c\| \leq \kappa_i N_{S_i}(c)$ for every $c \in A$; (G2) support closure, $L_i(\mathcal{A}(Z)) \subseteq \mathcal{A}(Z \cup S_i)$ for every region Z; (G3) contraction, $\|e^{tL_J}\| \leq 1$ for every finite set J of indices and every $t \geq 0$, where

$$L_J = \sum_{i \in J} L_i.$$

The family is determined by its data.
`Sealed.local_generators_iff`

7. The data derived from a family of local generators: the generator of a finite set J of indices is

$$L_J = \sum_{i \in J} L_i,$$

the full generator is $L = \sum_i L_i$, the dynamics is $t \mapsto e^{tL}$, the indices crossing a region Y are those i with S_i crossing Y, and the total strength is $K = \sum_i \kappa_i$. The dynamics does not increase norms: $\|e^{tL}a\| \leq \|a\|$ for $t \geq 0$.
`Sealed.generator_data`

Paths and the path-sum bound

Let X be the region of an observable and Y the region of a probe. The first-contact paths of length k from Y to X are sequences of indices $i_1, \ldots, i_k$; their weighted count $P_k(Y, X)$ enters the path-sum bound. For a proposition p, $[p]$ is 1 if p holds and 0 otherwise.

8. First-contact paths. There is no path of length 0. A sequence $i_1, \ldots, i_{k+1}$ of indices is a first-contact path from Y to X exactly when S_{i_1} crosses Y, each $S_{i_{j+1}}$ crosses S_{i_j} $(1 \leq j \leq k)$, the supports $S_{i_1}, \ldots, S_{i_k}$ are disjoint from X, and $S_{i_{k+1}}$ meets X.
`Sealed.first_contact_path`

9. The weighted path count.

$$P_k(Y, X) = \sum \kappa_{i_1} \cdots \kappa_{i_k},$$

the sum over the first-contact paths $i_1, \ldots, i_k$ of length k from Y to X. It satisfies the recursion $P_1(Y, X) = \sum \kappa_i$ over the indices i with S_i crossing Y and meeting X, and

$$P_{k+2}(Y, X) = \sum \kappa_i P_{k+1}(S_i, X)$$

over the indices i with S_i crossing Y and disjoint from X; here the second form of the count, defined by this recursion with value 0 at length 0, agrees with the first at every length $k \geq 1$. Every count is at most K^k, $K = \sum_i \kappa_i$.
`Sealed.path_count`

10. The terms of the series. The coefficient is $c_k(t) = (2t)^{k+1}/(k+1)!$; the k-th term of the path series is $c_k(t)P_{k+1}(Y,X)$. The commutator bound $C_{X,Y}(t)$ is the supremum of $\|[e^{tL}a, b]\|$ over a in the unit ball of $\mathcal{A}(X)$ and b in the unit ball of $\mathcal{A}(Y)$.
`Sealed.series_terms`

11. The path-sum bound. Let $\mathcal{A}$ be a net on the subsets of Λ that is causal for the lattice index, (S_i, L_i, κ_i) a family of local generators with $L = \sum_i L_i$, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ and $t \geq 0$. Then the path series

$$\sum_{k\geq 1} \frac{(2t)^k}{k!} P_k(Y,X)$$

converges and

$$\|[e^{tL}a, b]\| \leq 2\|a\|\|b\|\big([X \cap Y \neq \emptyset] + \sum_{k\geq 1} \frac{(2t)^k}{k!} P_k(Y,X)\big).$$

`Sealed.path_sum_bound`

12. The path-sum bound in supremum form: under the hypotheses of the path-sum bound, the commutator bound satisfies

$$C_{X,Y}(t) \leq 2\big([X \cap Y \neq \emptyset] + \sum_{k\geq 1} \frac{(2t)^k}{k!} P_k(Y,X)\big);$$

the series converges by the path-sum bound.
`Sealed.sup_bound`

Closed forms: the site matrix and walks

The touched sites are $T = \bigcup_i S_i$, a finite set. The site matrix

$$h_{uv} = \sum_{i:u,v\in S_i} \kappa_i$$

is a function on pairs of sites. The boundary of a region Y is the set ∂Y of sites of Y that lie in the support of an index crossing Y.

13. The site matrix and walk weights. The touched sites are $T = \bigcup_i S_i$, and

$$h_{uv} = \sum_i \kappa_i [u \in S_i][v \in S_i].$$

The walk weights are $w_0(y,x) = [y = x]$ and

$$w_{k+1}(y,x) = \sum_{z\in T} h_{yz} w_k(z,x).$$

The boundary ∂Y consists of the sites $y \in Y$ with $y \in S_i$ for some i with S_i crossing Y, and the touched sites of X are $X \cap T$. The walk sum is

$$W_k(Y,X) = \sum_{y\in\partial Y} \sum_{x\in X\cap T} w_k(y,x),$$

and the k-th term of the walk series is $\frac{(2t)^{k+1}}{(k+1)!} W_{k+1}(Y,X)$.
`Sealed.site_matrix_walks`

14. The walk bound. Under the hypotheses of the path-sum bound, the path count is dominated by the walk sum, $P_{k+1}(Y,X) \le W_{k+1}(Y,X)$ for every $k \ge 0$; the walk series

$$\sum_{k\ge 1} \frac{(2t)^k}{k!} W_k(Y,X)$$

converges; and

$$\|[e^{tL}a, b]\| \le 2\|a\|\|b\|\big([X \cap Y \ne \emptyset] + \sum_{k\ge 1} \frac{(2t)^k}{k!} W_k(Y,X)\big).$$

`Sealed.walk_bound`

15. The walk bound on a finite lattice. If the site set Λ is finite, the walk weights are the entries of the powers of the site matrix, $w_k(y,x) = (h^k)_{yx}$, and under the hypotheses of the path-sum bound

$$\|[e^{tL}a, b]\| \le 2\|a\|\|b\|\big([X \cap Y \ne \emptyset] + \sum_{y\in\partial Y} \sum_{x\in X\cap T} (e^{2th} - 1)_{yx}\big),$$

with e^{2th} the matrix exponential and 1 the identity matrix.
`Sealed.matrix_exponential_bound`

Closed forms: decay functions and bounded geometry

16. Decay kernels. Given $F : \Lambda \times \Lambda \to \mathbb{R}$ and $C \in \mathbb{R}$, there is a decay kernel with function F and reproducing constant $C_F = C$ exactly when $C > 0$, $F \ge 0$, and

$$\sum_{z\in T'} F(x,z)F(z,y) \le CF(x,y)$$

for all sites x, y and every finite set T' of sites. The kernel is determined by F and C.
`Sealed.decay_kernel_iff`

17. The decay-function bound. A real N is an F-bound of the family when $N \ge 0$ and $h_{xy} \le NF(x,y)$ for all sites x, y; the kernel sum is

$$\Sigma_F(Y,X) = \sum_{y\in\partial Y} \sum_{x\in X\cap T} F(y,x).$$

If N is an F-bound, then under the hypotheses of the path-sum bound

$$\|[e^{tL}a, b]\| \le 2\|a\|\|b\|\big([X \cap Y \ne \emptyset] + \frac{e^{2NC_F t} - 1}{C_F} \Sigma_F(Y,X)\big).$$

`Sealed.decay_bound`

18. Bounded geometry. Given $d : \Lambda \times \Lambda \to \mathbb{R}$, natural numbers g, R and reals ℓ, ρ, the family has bounded geometry with these data exactly when $d(u,u) = 0$, $d(u,w) \le d(u,v) + d(v,w)$, $\ell \ge 0$, $\rho \ge 0$, every site lies in the supports S_i of at most g indices i, every strength is at most ℓ, every support has at most R sites, and $d(u,v) \le \rho$ for all u, v in a common support. The bounded geometry is determined by these data; its row-sum bound is $M = g\ell R$ and its velocity $v = 2g\ell R$.
`Sealed.bounded_geometry_iff`

19. The bound under bounded geometry and the exponential light cone. Let the family have bounded geometry with distance d, diameter bound ρ and velocity $v = 2g\ell R$, let $r \geq 1$ be an integer, and let $d(y, x) > (r-1)\rho$ for every $y \in \partial Y$ and every $x \in X \cap T$. Under the hypotheses of the path-sum bound,

$$\|[e^{tL}a, b]\| \leq 2\|a\|\|b\|\big([X \cap Y \neq \emptyset] + |\partial Y|\frac{(vt)^r}{r!}e^{vt}\big),$$

and for every real μ,

$$\|[e^{tL}a, b]\| \leq 2\|a\|\|b\|\big([X \cap Y \neq \emptyset] + |\partial Y|e^{vt}e^{-\mu r}(e^{e^{\mu}vt} - 1)\big).$$

`Sealed.light_cone_bound`

Time reversal and the graded bracket

20. The negated family. The family is negation-contractive when $\|e^{-tL_J}\| \leq 1$ for every finite set J of indices and every $t \geq 0$. Then the negated family, with the same supports and strengths and the generators $-L_i$, is a family of local generators, and its dynamics is the dynamics of the family run backwards: $e^{t(-L)} = e^{-tL}$.

`Sealed.negated_family`

21. Time reversal. If the family is negation-contractive, then under the hypotheses of the path-sum bound with t any real number,

$$\|[e^{tL}a, b]\| \leq 2\|a\|\|b\|\big([X \cap Y \neq \emptyset] + \sum_{k \geq 1} \frac{(2|t|)^k}{k!}P_k(Y, X)\big)$$

(the series converges), and the bound also holds with the walk sums $W_k(Y, X)$ in place of $P_k(Y, X)$ (that series converges by the walk bound at time $|t|$).

`Sealed.time_reversal`

22. The graded bracket of $a, b \in A$ with freely chosen labels $|a|, |b| \in \{0, 1\}$ (no grading of A is assumed; 0 is even, 1 odd):

$$[a, b]_{\pm} = ab - (-1)^{|a||b|}ba.$$

It is the commutator $ab - ba$ when either element is even and the anticommutator $ab + ba$ when both are odd, and $\|[a, b]_{\pm}\| \leq 2\|a\|\|b\|$.

`Sealed.graded_bracket`

Hamiltonian families

Conventions: for a self-adjoint Hamiltonian H, the generator is $La = i[H, a]$ and the dynamics is $a \mapsto e^{itH}ae^{-itH}$.

23. Hamiltonian families. Given finite supports S_i and elements Φ_i of A for $i \in \iota$, there is a Hamiltonian family on $\mathcal{A}$ with these data exactly when every Φ_i lies in $\mathcal{A}(S_i)$ and is self-adjoint. The family is determined by its data.

`Sealed.interaction_iff`

24. The Hamiltonian family as local generators. For $x \in A$ the derivation is $\mathrm{ad}_x(a) = i[x, a]$. For a Hamiltonian family (S_i, Φ_i) in a nonzero C*-algebra, with

$$H_J = \sum_{i \in J} \Phi_i$$

and $H = H_\iota$ the full Hamiltonian: the family of local generators of (S_i, Φ_i) has supports S_i, generators $L_i = \mathrm{ad}_{\Phi_i}$ and strengths $\kappa_i = \|\Phi_i\|$; it is negation-contractive; $U_t = e^{itH}$ is unitary; and the dynamics is the conjugation $e^{tL}a = U_t a U_t^*$.
`Sealed.hamiltonian_generators`

Hamiltonian dynamics: the hypotheses and the role exchange

Let $\mathcal{A}$ be a net over the subsets of a set Λ of sites in a C*-algebra A. The derivation of $x \in A$ is $\mathrm{ad}_x a = i[x, a]$. A Hamiltonian family $(S_i, \Phi_i)_{i \in I}$, indexed by a finite set I, has finite supports $S_i \subseteq \Lambda$ and self-adjoint terms $\Phi_i \in \mathcal{A}(S_i)$; its Hamiltonian is $H = \sum_i \Phi_i$, $U_t = \exp(itH)$, its local generators are $(S_i, \mathrm{ad}_{\Phi_i}, \|\Phi_i\|)$, and $A(t) = e^{tL}a$ with

$$L = \sum_i \mathrm{ad}_{\Phi_i}.$$

Norms of maps of A are operator norms. For regions X, Y, $[X \cap Y \neq \emptyset]$ is 1 if X and Y meet and 0 otherwise.

25. Hamiltonian families satisfy the three hypotheses of the path-sum bound, the third by unitarity. Let A be nontrivial. For $x \in \mathcal{A}(S)$ and every $c \in A$, $\|\mathrm{ad}_x c\| \leq \|x\| N_S(c)$ (locality in the argument with strength $\|x\|$), and $\mathrm{ad}_x c \in \mathcal{A}(Z \cup S)$ whenever $c \in \mathcal{A}(Z)$ (support closure). For a Hamiltonian family $(S_i, \Phi_i)_{i \in I}$, every sub-family $J \subseteq I$ and every real t, of either sign,

$$\| \exp(t \sum_{i \in J} \mathrm{ad}_{\Phi_i}) \| \leq 1$$

(contraction); for every real t, $U_t^* = U_{-t}$, and $\|A(t)\| = \|a\|$ for every $a \in A$.
`Sealed.spin_hamiltonian_hypotheses`

26. Role exchange ([Nachtergaele–Sims–Young 2019], eqs. (3.24) and (3.39)). Let A be nontrivial. For a unitary $U \in A$ and $a, b \in A$,

$$\|[UaU^*, b]\| = \|[a, U^*bU]\|.$$

For a Hamiltonian family, every real t and $a, b \in A$,

$$\|[A(t), b]\| = \|[B(-t), a]\|,$$

where $B(s) = e^{sL}b$. If moreover the net is causal for the lattice causal index, $a \in \mathcal{A}(X)$ and $b \in \mathcal{A}(Y)$, then for every real t the walk series of the pairs (Y, X) and (X, Y) at time $|t|$ converge and

$$\|[A(t), b]\| \leq 2\|a\|\|b\| \Big([X \cap Y \neq \emptyset] + \min \Big\{ \sum_{k \geq 0} \omega_k(Y, X; |t|), \sum_{k \geq 0} \omega_k(X, Y; |t|) \Big\} \Big),$$

where

$$\omega_k(Y, X; s) = \frac{(2s)^{k+1}}{(k+1)!} W_{k+1}(Y, X)$$

are the walk terms of the pair (Y, X) for the strengths $\|\Phi_i\|$, with ∂ the boundaries and T the touched sites of the family; and for every decay kernel (F, C_F) and every F-bound N of the family,

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big([X \cap Y \neq \emptyset] + \frac{e^{2NC_F|t|} - 1}{C_F} \min\{\Sigma_F(Y, X), \Sigma_F(X, Y)\}\Big),$$

where

$$\Sigma_F(Y, X) = \sum_{y \in \partial Y} \sum_{x \in X \cap T} F(y, x).$$

`Sealed.spin_role_exchange`

Qudit lattices

Let Λ be a finite set of sites, each site u carrying a finite alphabet Ω_u. A configuration is a choice $s = (s(u))_{u \in \Lambda}$ of a letter at each site. For a region $X \subseteq \Lambda$ a matrix on the letters of X is a complex matrix indexed by the choices of letters at the sites of X, and $s|_X$ is the restriction of s to X.

27. The qudit algebra and its local algebras. The qudit algebra is the algebra of complex matrices indexed by the configurations, with the operator norm of the matrix acting on the Euclidean space of complex functions on the configurations. The embedding of a matrix m on the letters of a region X is the matrix $\iota_X m$ with

$$(\iota_X m)_{ss'} = m_{s|_X, s'|_X}$$

if s and s' agree at every site outside X, and 0 otherwise. A matrix M lies in the local algebra $\mathcal{A}(X)$ of the qudit net exactly when (1) $M_{ss'} = 0$ whenever s and s' differ at some site outside X, and (2) $M_{ss'} = M_{rr'}$ whenever s, s' agree outside X, r, r' agree outside X, $s = r$ on X and $s' = r'$ on X. $\mathcal{A}(\Lambda)$ is the whole qudit algebra, and the qudit net is causal for the lattice causal index. If every alphabet is nonempty, $\mathcal{A}(X)$ consists exactly of the embeddings $\iota_X m$.
`Sealed.spin_qudit_net`

28. Single-site operators. Let every site carry the same finite alphabet Ω. For a site u and a matrix m on Ω, the single-site operator m_u has the entries

$$(m_u)_{ss'} = m_{s(u), s'(u)}$$

if s and s' agree at every site other than u, and 0 otherwise. It lies in $\mathcal{A}(\{u\})$; $m_u m'_u = (mm')_u$ and $(m_u)^* = (m^*)_u$; and for sites $u \neq v$, m_u and m'_v commute, and

$$(m_u m'_v)_{ss''} = m_{s(u), s''(u)} m'_{s(v), s''(v)}$$

whenever s and s'' agree at every site other than u and v.
`Sealed.spin_siteOp`

29. Qudit interactions. Let every alphabet be finite and nonempty. A qudit interaction indexed by a finite set I consists of finite supports $S_i \subseteq \Lambda$ and Hermitian matrices m_i on the letters of S_i; for all such supports and matrices there is a qudit interaction with the supports S_i whose Hamiltonian family has the terms $\Phi_i = \iota_{S_i} m_i$. For every qudit interaction the matrices are Hermitian, its Hamiltonian family on the qudit net has the same supports and the terms $\Phi_i = \iota_{S_i} m_i$, and its family of local generators is that of its Hamiltonian family.
`Sealed.spin_quditInteraction`

The published bounds for spin systems

Let Λ be a finite set of sites with finite nonempty alphabets, and $(S_i, m_i)_{i\in I}$ a qudit interaction with the terms $\Phi_i = \iota_{S_i} m_i$, the strengths $\kappa_i = \|\Phi_i\|$, the touched sites T, the boundaries ∂Y and the site matrix h; $A(t) = e^{tL}a$ is its dynamics.

30. The spin-system Lieb–Robinson bound in matrix form (the bound of [Chen–Lucas–Yin 2023], eq. (3.36), with the identity subtracted and the sum restricted to the boundary and the touched sites). Here $e^{2|t|h}$ is the exponential of the site matrix, regarded as the real $\Lambda \times \Lambda$ matrix with entries h_{uv}, and 1 is the identity matrix. For $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ and every real t,

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big([X \cap Y \ne \emptyset] + \min\Big\{ \sum_{y\in\partial Y} \sum_{x\in X\cap T} (e^{2|t|h} - 1)_{yx}, \sum_{x\in\partial X} \sum_{y\in Y\cap T} (e^{2|t|h} - 1)_{xy} \Big\}\Big).$$

`Sealed.spin_qudit_bound_matrix`

31. The spin-system Lieb–Robinson bound in decay form (Theorem 3.1 of [Nachtergaele–Sims–Young 2019] for time-independent interactions). For a decay function (F, C_F) and an F-bound N of the interaction, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ and every real t,

$$\|[A(t), b]\| \le 2\|a\|\|b\|\Big([X \cap Y \ne \emptyset] + \frac{e^{2NC_F|t|} - 1}{C_F} \min\{\Sigma_F(Y, X), \Sigma_F(X, Y)\}\Big),$$

with

$$\Sigma_F(Y, X) = \sum_{y\in\partial Y} \sum_{x\in X\cap T} F(y, x).$$

If Λ is nonempty and $F > 0$, the F-norm

$$\|\Phi\|_F = \max_{x,y} h_{xy}/F(x, y)$$

(their eq. (3.14)) is an F-bound and the least one. For a function d on pairs of sites with $d(x, y) \le d(x, z) + d(z, y)$ and $\mu \ge 0$, the weighted function

$$F_\mu(x, y) = e^{-\mu d(x,y)} F(x, y)$$

is a decay kernel with the same constant C_F (their eq. (3.11)).

`Sealed.spin_qudit_bound_decay`

32. The spin-system Lieb–Robinson bound under bounded geometry, with the light cone. Let the interaction have bounded geometry with the distance d, the diameter bound ρ and the velocity v, let $r \ge 1$ be an integer with $(r - 1)\rho < d(y, x)$ for all $y \in \partial Y$ and $x \in X \cap T$, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$ and $t \ge 0$. Then

$$\|[A(t), b]\| \le 2\|a\|\|b\|([X \cap Y \ne \emptyset] + |\partial Y| \frac{(vt)^r}{r!} e^{vt}),$$

and for every real μ,

$$\|[A(t), b]\| \le 2\|a\|\|b\|([X \cap Y \ne \emptyset] + |\partial Y| e^{vt} e^{-\mu r}(e^{e^\mu vt} - 1)).$$

`Sealed.spin_qudit_bound_geometry`

The conditional expectation and the twirl

Let Λ be a finite set of sites with finite nonempty alphabets. For a region X, Ω_{X^c} is the set of choices η of letters at the sites outside X, and $(s|_X, \eta)$ is the configuration equal to s on X and to η outside X. For the twirl each alphabet is cyclic, $\Omega_u = \mathbb{Z}_{q_u}$ with $q_u \geq 1$, letters are added modulo q_u, and $e_n(k) = e^{2\pi ik/n}$, computed as $(e^{2\pi i/n})^{\bar{k}}$ with $\bar{k} \in \{0, \ldots, n-1\}$ the representative of $k \in \mathbb{Z}_n$.

33. The conditional expectation onto a region X is the map P_X of the qudit algebra with

$$(P_X M)_{ss'} = \frac{1}{|\Omega_{X^c}|} \sum_{\eta \in \Omega_{X^c}} M_{(s|_X,\eta),(s'|_X,\eta)}$$

if s and s' agree at every site outside X, and $(P_X M)_{ss'} = 0$ otherwise. It maps into $\mathcal{A}(X)$, is additive and complex homogeneous, commutes with the adjoint, satisfies $P_X M = M$ for $M \in \mathcal{A}(X)$ and $P_X 1 = 1$; $P_\Lambda M = M$, and $P_\emptyset M = (\operatorname{tr} M/|\Omega_\Lambda|)1$, with $|\Omega_\Lambda|$ the number of configurations.
`Sealed.spin_condExp`

34. Weyl unitaries and the twirl, for cyclic alphabets. For a region Y and choices ξ, η of letters at the sites of Y, the Weyl unitary $V_{\xi,\eta}$ has the entries $(V_{\xi,\eta})_{ss'} = \psi_\eta(s')$ if $s = s' + \xi$ (that is, $s(u) = s'(u) + \xi(u)$ for $u \in Y$ and $s(u) = s'(u)$ for $u \notin Y$), and 0 otherwise, where

$$\psi_\eta(s) = \prod_{u \in Y} e_{q_u}(\eta(u)s(u)).$$

It is unitary and lies in $\mathcal{A}(Y)$. For every region X and every matrix M the twirl onto X, $|\Omega_{X^c}|^{-2} \sum_{\xi,\eta \in \Omega_{X^c}} V_{\xi,\eta} M V_{\xi,\eta}^*$ over the Weyl unitaries of the complement X^c, equals $P_X M$; hence $\|P_X M\| \leq \|M\|$, and $P_X(aM) = aP_X M$ and $P_X(Mb) = (P_X M)b$ for $a, b \in \mathcal{A}(X)$.
`Sealed.spin_twirl`

Local approximation

Let $\mathcal{A}$ be a net over the subsets of a set Λ of sites in a C*-algebra A, and N_S its commutator seminorms.

35. Nets with local approximation and approximate locality. The net has local approximation with the maps $P_X : A \to A$ exactly when $P_X c \in \mathcal{A}(X)$ and $\|c - P_X c\| \leq N_{X^c}(c)$ for every region X and every $c \in A$. If the net is causal for the lattice causal index, $N_Y(c) \leq 2\|c - c'\|$ for every $c \in A$ whenever $c' \in \mathcal{A}(X)$ and Y is disjoint from X. A map $\mathcal{T} : A \to A$ is ε-local with the neighbourhood map $\mathcal{N}$ (a map from regions to regions) exactly when for every region O and every $c \in \mathcal{A}(O)$ there is $c' \in \mathcal{A}(\mathcal{N}(O))$ with $\|\mathcal{T}c - c'\| \leq \varepsilon\|c\|$; it is 0-local exactly when $\mathcal{T}(\mathcal{A}(O)) \subseteq \mathcal{A}(\mathcal{N}(O))$ for every region O.
`Sealed.spin_localApprox_iff`

36. Local approximation on the qudit lattice. For a finite set Λ of sites with cyclic alphabets, every region X and every matrix M,

$$\|M - P_X M\| \leq N_{X^c}(M) \leq 2\|M - P_X M\|,$$

where N_{X^c} is the commutator seminorm of the qudit net for the complement X^c and P_X the conditional expectation onto X. The conditional expectations give the qudit net with cyclic alphabets local approximation.
`Sealed.spin_qudit_localApprox`

Truncated dynamics and the approximate automaton

Let $(S_i, L_i, \kappa_i)_{i\in I}$ be a family of local generators on a net over the subsets of a set Λ of sites in a C*-algebra A, causal for the lattice causal index, with local approximation P; $T = \bigcup_i S_i$ is the set of touched sites, $A(t) = e^{tL}a$, and

$$\omega_k(Y, X; t) = \frac{(2t)^{k+1}}{(k+1)!} W_{k+1}(Y, X)$$

are the walk terms of a pair of regions (Y, X), with ∂ the boundaries of the family; termwise

$$\omega_{k-1}(Y, X; t) = \frac{(2t)^k}{k!} W_k(Y, X),$$

so that

$$\sum_{k\geq 0} \omega_k(Y, X; t) = \sum_{k\geq 1} \frac{(2t)^k}{k!} W_k(Y, X)$$

is the walk series of the walk bound. A map $\mathcal{T} : A \to A$ is ε-local with a neighbourhood map $\mathcal{N}$ (a map from regions to regions) if for every region O and every $c \in \mathcal{A}(O)$ there is $c' \in \mathcal{A}(\mathcal{N}(O))$ with $\|\mathcal{T}c - c'\| \leq \varepsilon\|c\|$. Under bounded geometry with the distance d and the diameter bound ρ, the r-neighbourhood of a region O, for an integer r, is

$$O^{+r} = \{w : d(w, x) < r\rho \text{ for some } x \in O\}.$$

37. Truncated dynamics. For $a \in \mathcal{A}(X)$, a region $X' \supseteq X$, $Y = X'^c$ and $t \geq 0$, the walk series of the pair (Y, X) converges, and

$$\|A(t) - P_{X'}A(t)\| \leq 2\|a\| \sum_{k\geq 0} \omega_k(Y, X; t)$$

and

$$N_Y(A(t)) \leq 2\|a\| \sum_{k\geq 0} \omega_k(Y, X; t).$$

Under bounded geometry with the distance d, the diameter bound ρ and the velocity v, if $r \geq 1$ is an integer with $(r-1)\rho < d(y, x)$ for all $y \in \partial Y$ and $x \in X \cap T$, then

$$\|A(t) - P_{X'}A(t)\| \leq 2\|a\||\partial Y| \frac{(vt)^r}{r!} e^{vt}$$

and, for every real μ,

$$\|A(t) - P_{X'}A(t)\| \leq 2\|a\||\partial Y| e^{vt} e^{-\mu r}(e^{e^{\mu} vt} - 1).$$

`Sealed.spin_truncated_dynamics`

38. The approximate automaton. Let $t \geq 0$; the r-neighbourhoods are as defined above. (a) Let Λ be finite and $\mathcal{N}$ a neighbourhood map. For every region O the series

$$\sum_{k\geq 0} \omega_k(\mathcal{N}(O)^c, O; t)$$

converges; let $\varepsilon(t)$ be the largest, over the finitely many regions O, of

$$2\sum_{k\geq 0} \omega_k(\mathcal{N}(O)^c, O; t).$$

If $O \subseteq \mathcal{N}(O)$ for every region O, then e^{tL} is $\varepsilon(t)$-local with $\mathcal{N}$. (b) Under bounded geometry with the diameter bound $\rho > 0$ and the velocity v, and for an integer $r \geq 1$, e^{tL} is $2|T|\frac{(vt)^r}{r!}e^{vt}$-local with $O \mapsto O^{+r}$, and for every real μ it is $2|T|e^{vt}e^{-\mu r}(e^{e^{\mu} vt} - 1)$-local with $O \mapsto O^{+r}$.

`Sealed.spin_approx_automaton`

39. Truncated dynamics and the approximate automaton for spin systems. Let Λ be a finite set of sites with cyclic alphabets, $(S_i, m_i)_{i\in I}$ a qudit interaction with the strengths $\kappa_i = \|\Phi_i\|$ and the touched sites T, P the conditional expectations and $t \geq 0$. For $a \in \mathcal{A}(X)$ and a region $X' \supseteq X$ the walk series of the pair (X'^c, X) converges and

$$\|A(t) - P_{X'}A(t)\| \leq 2\|a\| \sum_{k\geq 0} \omega_k(X'^c, X; t).$$

Under bounded geometry with the distance d, the diameter bound $\rho > 0$ and the velocity v, and for an integer $r \geq 1$, the dynamics e^{tL} is $2|T|\frac{(vt)^r}{r!}e^{vt}$-local and, for every real μ, $2|T|e^{vt}e^{-\mu r}(e^{e^\mu vt}-1)$-local with the r-neighbourhood map $O \mapsto O^{+r}$.

`Sealed.spin_qudit_automaton`

Open systems

Let $\mathcal{A}$ be a net over the subsets of a set Λ of sites in a C*-algebra A. For $\Gamma \in A$ the dissipator is

$$\Delta_\Gamma a = \Gamma^* a \Gamma - \frac{1}{2}(\Gamma^*\Gamma a + a\Gamma^*\Gamma).$$

A Lindblad term on a finite region S consists of a self-adjoint $H \in \mathcal{A}(S)$ (its Hamiltonian part) and finitely many jump operators $\Gamma_1, \ldots, \Gamma_m \in \mathcal{A}(S)$; its generator is

$$L_S a = i[H, a] + \sum_j \Delta_{\Gamma_j} a.$$

Norms of maps of A are operator norms.

40. Lindblad terms. For a finite region S, a self-adjoint $H \in \mathcal{A}(S)$ and $\Gamma_1, \ldots, \Gamma_m \in \mathcal{A}(S)$ there is a Lindblad term with support S, Hamiltonian part H and the generator

$$L_S a = i[H, a] + \sum_j \Delta_{\Gamma_j} a$$

(only the generator is fixed: the term's jump operators are not asserted to be $\Gamma_1, \ldots, \Gamma_m$, and the statements below hold for its own jump operators). For every Lindblad term with support S, Hamiltonian part H and jump operators Γ_j: H is self-adjoint, H and the Γ_j lie in $\mathcal{A}(S)$, and the generator has this form; it satisfies

$$\|L_S\| \leq 2\|H\| + 2\sum_j \|\Gamma_j\|^2,$$

maps $\mathcal{A}(Z)$ into $\mathcal{A}(Z \cup S)$, satisfies $(L_S a)^* = L_S(a^*)$ and

$$\|L_S c\| \leq (\|H\| + \sum_j \|\Gamma_j\|^2) N_S(c)$$

for every c. If the net is causal for the lattice causal index, $L_S c = 0$ for $c \in \mathcal{A}(S^c)$, and for every local approximation P of the net and every c, $L_S c = L_S(c - P_{S^c}c)$ and $\|L_S c\| \leq \|L_S\| N_S(c)$.

`Sealed.spin_lindbladTerm`

41. Open systems as local generators. A Lindblad family indexed by a finite set I is a family $(T_i)_{i\in I}$ of Lindblad terms, with supports S_i, Hamiltonian parts H_i, jump operators Γ_{ij} and generators L_i, such that every sub-family is contractive:

$$\|\exp(t\sum_{i\in J} L_i)\| \le 1$$

for every $J \subseteq I$ and $t \ge 0$. For a Lindblad family, its family of local generators with the direct strengths has the supports S_i, the generators L_i and the strengths

$$\kappa_i = \|H_i\| + \sum_j \|\Gamma_{ij}\|^2.$$

Let the net be causal for the lattice causal index, $a \in \mathcal{A}(X)$, $b \in \mathcal{A}(Y)$, $t \ge 0$ and $A(t) = e^{tL}a$ with $L = \sum_i L_i$. With the direct strengths the walk series of the pair (Y, X) converges and

$$\|[A(t), b]\| \le 2\|a\|\|b\|([X \cap Y \ne \emptyset] + \sum_{k\ge 0} \omega_k(Y, X; t)).$$

For every local approximation P of the net, the family of local generators with the strengths $\kappa_i = \|L_i\|$ has the supports S_i and the generators L_i; with these strengths the walk series of the pair (Y, X) converges, the same walk bound holds, and

$$\|A(t) - P_{X'}A(t)\| \le 2\|a\| \sum_{k\ge 0} \omega_k(X'^c, X; t)$$

for every region $X' \supseteq X$, this series converging as well.

`Sealed.spin_lindbladFamily`

42. A dephasing term is contractive. Let A be a nontrivial C*-algebra and $\Gamma \in A$ self-adjoint with $\Gamma^2 = 1$. For every real t and every $a \in A$,

$$e^{t\Delta_\Gamma}a = \frac{1 + e^{-2t}}{2}a + \frac{1 - e^{-2t}}{2}\Gamma a \Gamma,$$

and $\|e^{t\Delta_\Gamma}\| \le 1$ for $t \ge 0$.

`Sealed.spin_dephasing`

Harmonic bosons: the lattice phase space and the bosonic net

Throughout this section Λ is a finite set of sites. A phase-space vector is a real function f on the coordinates $\Lambda \times \{0, 1\}$: $f(x, 0) = f^q_x$ is the position and $f(x, 1) = f^p_x$ the momentum coordinate at the site x. The phase space V is the real vector space of these functions, and $\mathrm{CCR}(V, \sigma)$ is the CCR algebra of the symplectic form σ below, with its Weyl operators $W(f)$; its norm is the C*-norm.

43. The lattice phase space. For a finite set Λ of sites the symplectic form is

$$\sigma(f, g) = \sum_{x\in\Lambda}(f^q_x g^p_x - f^p_x g^q_x),$$

and $\sigma(f, g) = f \cdot \Omega g$ (dot product with the matrix-vector product), where the real matrix Ω on $\Lambda \times \{0, 1\}$ has the entries $\Omega_{(x,0),(x,1)} = 1$ and $\Omega_{(x,1),(x,0)} = -1$ for every site x, all other entries being 0. A phase-space vector f is supported in a set X of sites when $f(x, 0) = f(x, 1) = 0$ for every site $x \notin X$.

`Sealed.bos_phase_space`

44. The form and the bosonic net. The symplectic form σ is alternating, $\sigma(f,f) = 0$, and $\sigma(f,g) = 0$ whenever f is supported in X, g is supported in Y and the sets of sites X and Y are disjoint. The bosonic net assigns to a set X of sites the norm closure $\mathcal{A}(X)$ of the star subalgebra of $\mathrm{CCR}(V,\sigma)$ generated by the Weyl operators $W(f)$ of the phase-space vectors f supported in X; it is causal for the lattice causal index: for disjoint X and Y, every element of $\mathcal{A}(X)$ commutes with every element of $\mathcal{A}(Y)$.

`Sealed.bos_weylNet`

The Weyl commutator

45. The Weyl commutator. For all phase-space vectors f and g, the Weyl operators of $\mathrm{CCR}(V,\sigma)$ satisfy

$$[W(f),W(g)] = (e^{-i\sigma(f,g)} - 1)\, W(g)W(f),$$

and

$$\|[W(f),W(g)]\| = 2\,|\sin(\sigma(f,g)/2)| \le |\sigma(f,g)|.$$

`Sealed.bos_weyl_commutator`

Quadratic generators, the flow and the dynamics

A real matrix on $\Lambda \times \{0,1\}$ acts on phase-space vectors by the matrix-vector product,

$$(Kf)_b = \sum_a K_{ba} f_a.$$

For a real matrix M over a finite index set,

$$e^M = \sum_{n \ge 0} M^n/n!$$

is the matrix exponential.

46. Quadratic generators. A real matrix K on $\Lambda \times \{0,1\}$ is a quadratic generator exactly when $K^{\mathsf{T}}\Omega + \Omega K = 0$, and this holds exactly when $\sigma(Ku,v) + \sigma(u,Kv) = 0$ for all phase-space vectors u and v.

`Sealed.bos_isQuadraticGenerator_iff`

47. The flow. For a real matrix K on $\Lambda \times \{0,1\}$ the flow is the matrix exponential $T_t = e^{tK}$, $t \in \mathbb{R}$; it satisfies $T_0 = 1$ and $T_{s+t} = T_s T_t$ for all real s, t. If K is a quadratic generator, every T_t is symplectic: $T_t^{\mathsf{T}}\Omega T_t = \Omega$, and $\sigma(T_t u, T_t v) = \sigma(u,v)$ for all phase-space vectors u, v.

`Sealed.bos_flow`

48. The dynamics. Let K be a quadratic generator with flow T_t. For every real t the dynamics α_t is a ($\mathbb{C}$-linear) star automorphism of $\mathrm{CCR}(V,\sigma)$ with $\alpha_t(W(f)) = W(T_t f)$ for every phase-space vector f, and it is the only star automorphism with this property. Moreover α_0 is the identity and $\alpha_{s+t} = \alpha_s \circ \alpha_t$ for all real s, t.

`Sealed.bos_alpha`

The entrywise bound

For a real matrix K over a finite index set ι, $|K|$ is the matrix of absolute values, $|K|_{ba} = |K_{ba}|$.

49. The entrywise exponential bound. Let K be a real matrix over a finite index set ι and $t \in \mathbb{R}$. The entries of the matrix exponential are the sums of convergent series,

$$(e^{tK})_{ba} = \sum_{n \geq 0} t^n (K^n)_{ba}/n!,$$

and

$$|(e^{tK})_{ba}| \leq (e^{|t|\,|K|})_{ba}$$

for all indices b, a.

`Sealed.bos_abs_exp_apply_le`

50. Range and row sums. Let K be a real matrix over a finite index set ι, $s : \iota \to \Lambda$ a map to a set of sites, d a real function of two sites and $R, v \in \mathbb{R}$. The matrix K has range R for s and d exactly when $K_{ba} = 0$ for all indices b, a with $d(s(b), s(a)) > R$, and it has row sums at most v exactly when $\sum_a |K_{ba}| \leq v$ for every index b.

`Sealed.bos_hasRange_rowSumLE_iff`

The harmonic Lieb–Robinson bound

For a finite set X of sites, $\hat{X} = X \times \{0, 1\}$ is the set of its coordinates, and

$$\|f\|_1 = \sum_a |f_a|$$

is the sum of the absolute values of all coordinates of f. The site of a coordinate (x, i) is x.

51. The harmonic Lieb–Robinson bound. Let K be a quadratic generator with flow T_t and dynamics α_t, let X and Y be finite sets of sites, f a phase-space vector supported in X, g one supported in Y, and $t \in \mathbb{R}$. For a coordinate $b = (y, i)$ let $\bar{b}$ be the other coordinate at the same site: $(y, 1)$ for $i = 0$ and $(y, 0)$ for $i = 1$. Then

$$\|[\alpha_t(W(f)), W(g)]\| \leq |\sigma(T_t f, g)| \leq \sum_{b \in \hat{Y}} \sum_{a \in \hat{X}} (e^{|t|\,|K|})_{ba}\, |f_a|\, |g_{\bar{b}}|.$$

`Sealed.bos_liebRobinson`

52. The harmonic bound, range form. Let d be a real function of two sites with $d(x, x) = 0$ and $d(x, z) \leq d(x, y) + d(y, z)$ for all sites, and let K be a quadratic generator with dynamics α_t that has range R for the site map and d and row sums at most $v \geq 0$. Let f be supported in a finite set X of sites, g in a finite set Y, $t \geq 0$, and r a natural number with $nR < d(y, x)$ for all $y \in Y$, $x \in X$ and natural numbers $n < r$ (for $r \geq 1$ the case $n = 0$ requires $d(y, x) > 0$, so that X and Y are disjoint; for $r = 0$ the condition is empty). Then

$$\|[\alpha_t(W(f)), W(g)]\| \leq \|f\|_1 \|g\|_1 \, \frac{(vt)^r}{r!}\, e^{vt}.$$

`Sealed.bos_liebRobinson_range`

53. The harmonic bound, light-cone form. Under the hypotheses of the range form (d a real function of two sites with $d(x, x) = 0$ and the triangle inequality, K a quadratic generator of range R and row sums at most $v \geq 0$, f supported in a finite set X of sites, g in a finite set Y, $t \geq 0$, and $nR < d(y, x)$ for all $y \in Y$, $x \in X$, $n < r$), and for $\mu \geq 0$ and $r \geq 1$,

$$\|[\alpha_t(W(f)), W(g)]\| \leq \|f\|_1 \|g\|_1 \, e^{-\mu r}(e^{e^{\mu} vt} - 1).$$

`Sealed.bos_liebRobinson_cone`

The harmonic chain

The chain has the sites $\{0, \dots, n-1\}$; two sites x, y are neighbours when $x+1 = y$ or $y+1 = x$, and $\deg x$ is the number of neighbours of x.

54. The harmonic chain. Let n be a natural number and $\omega, \lambda \in \mathbb{R}$. The potential matrix A of the chain is the real $n \times n$ matrix with $A_{xx} = \omega^2 + \lambda \deg x$, $A_{xy} = -\lambda$ for neighbours x, y, and $A_{xy} = 0$ otherwise. The chain generator K is the real matrix on $\{0, \dots, n-1\} \times \{0, 1\}$ with $K_{(x,0),(y,1)} = -A_{xy}$, $K_{(x,1),(x,0)} = 1$ and all other entries 0, so that $(Kf)^q = -Af^p$ and $(Kf)^p = f^q$. The matrix K is a quadratic generator; it has range 1 for the site map and the distance $d(x, y) = |x - y|$; and for $\lambda \geq 0$ its row sums are at most $\max(1, \omega^2 + 4\lambda)$.

`Sealed.bos_chain`

55. The bound for the harmonic chain. Let K be the chain generator on n sites with $\omega \in \mathbb{R}$ and $\lambda \geq 0$, assumed to be a quadratic generator (which it is for all ω, λ), α_t its dynamics and $v = \max(1, \omega^2 + 4\lambda)$. Let f be supported in a finite set X of sites, g in a finite set Y, $t \geq 0$, and r a natural number with $r \leq |y - x|$ for all $y \in Y$ and $x \in X$. Then

$$\|[\alpha_t(W(f)), W(g)]\| \leq \|f\|_1 \|g\|_1 \, \frac{(vt)^r}{r!} \, e^{vt},$$

and for $\mu \geq 0$ and $r \geq 1$ also

$$\|[\alpha_t(W(f)), W(g)]\| \leq \|f\|_1 \|g\|_1 \, e^{-\mu r} (e^{e^{\mu} vt} - 1).$$

`Sealed.bos_chain_bound`

Examples

These statements show that the hypotheses of the theorems above can be met and that the objects are not degenerate. The two-qubit net is the qudit net with sites $\Lambda = \{0, 1\}$ and the alphabet $\{0, 1\}$ at each site;

$$\sigma^x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

and

$$\sigma^z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

and σ^x_u, σ^z_u are their single-site operators at the site u.

56. On a finite nonempty site set, the constant function $F = 1$ with reproducing constant $C_F = |\Lambda|$ is a decay kernel, and for every family of local generators the total strength $K = \sum_i \kappa_i$ is an F-bound for it.

`Sealed.witness_decay_kernel`

57. The qudit net and the spin dynamics are not trivial. On two qubits, σ^x_1 does not lie in $\mathcal{A}(\{0\})$, and σ^x_0, σ^z_0 do not commute. There is a Hamiltonian family on the two-qubit net with a single term, of support $\{0, 1\}$, equal to $H = \sigma^x_0 \sigma^x_1$; for it $a = \sigma^z_0 \in \mathcal{A}(\{0\})$ and $b = \sigma^z_1 \in \mathcal{A}(\{1\})$ commute, while $[A(t), b] \neq 0$ for some $t > 0$.

`Sealed.witness_spin_two_qubit`

58. Open systems on two qubits. There is a Lindblad family on the two-qubit net with a single term, of support $\{0\}$, Hamiltonian part 0 and generator $La = \sigma_0^z a \sigma_0^z - a$; so the contraction hypothesis of a Lindblad family can be met. There is a Lindblad term of support $\{0, 1\}$ with Hamiltonian part $H = \sigma_0^x \sigma_1^x$ and generator

$$La = i[H, a] + \sigma_0^z a \sigma_0^z - a$$

(that is, $i[H, a] + \Delta_{\sigma_0^z} a$) such that $[e^{tL}\sigma_0^z, \sigma_1^z] \neq 0$ for some $t > 0$.
`Sealed.witness_spin_open_system`

59. Two Weyl operators that do not commute. On a single site, for the phase-space vectors f with $f^q = \pi$, $f^p = 0$ and g with $g^q = 0$, $g^p = 1$, one has $\sigma(f, g) = \pi$ and $\|[W(f), W(g)]\| = 2$.
`Sealed.witness_bos_weyl_noncommuting`

60. Coupled momenta in the harmonic chain. Let K be the chain generator on $n + 2$ sites with $\omega \in \mathbb{R}$ and coupling $\lambda > 0$, assumed to be a quadratic generator (which it is), and α_t its dynamics. For the momentum unit vectors $f = \delta_{(0,1)}$ at the site 0 and $g = \delta_{(1,1)}$ at the site 1 there is a time $t > 0$ with $[\alpha_t(W(f)), W(g)] \neq 0$.
`Sealed.witness_bos_chain_noncommuting`

SHA-256 of `verify/Challenge.lean`: `077a306afab7894be1fa157bf738e140e70bca3d3ed78b91f579d8676d6c3a10`